Sahaj Bhadja

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Summary

Associate software engineer with 2+ years building **full-stack applications**, **tested backend services**, and **iterative Agile deliveries** with strong focus on craftsmanship, clear communication, and dependable team execution.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **full-stack workflow services** used by 500+ users, reducing latency from **1.2s to 350ms** through SQL tuning and backend optimization.
- Implemented **automated validation and monitoring checks** for 5K+ monthly runs, improving production stability and reducing risk in downstream systems.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **user-facing product features** handling 10K+ daily interactions, translating requirements into maintainable APIs and responsive web interfaces.
- Improved **deployment and QA readiness** via CI/CD checks and defect triage, reducing issue turnaround by **60%** in Agile sprint cycles.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **ML and analytics pipelines** over 100K+ records, enabling reproducible evaluation and structured experimentation workflows.
- Produced **developer documentation and test scenarios** pre-release, improving handoff quality and reducing implementation ambiguity across teams.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API and ETL services** with cloud-backed workflows, improving response times by **30%** under production workloads.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito, improving CI pipeline confidence by **68%** and reducing regressions.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Developed **web UI and integration modules** for 2M+ synchronized records across 8 workflows, improving responsiveness of event-driven operations.
- Automated **Docker-based build and test flows** with Jenkins, reducing release time by **50%** and improving deployment consistency.

Skills

Languages: **Python**, **JavaScript/TypeScript**, SQL, Java, C#, Bash
Application Engineering: React/HTML/CSS UI development, RESTful APIs, backend service design, integration workflows
Data: relational databases (SQL Server/MySQL/PostgreSQL), query optimization, JSON/CSV processing
Operations: Git/GitHub, pull requests, code reviews, CI/CD pipelines, deployment validation, production support basics
Engineering Practice: Agile/Scrum, estimation and planning, debugging/testing, concise technical documentation

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built a **schema-aware NL-to-SQL assistant** across 10+ datasets with validation and explainable outputs, reducing turnaround for ad-hoc data analysis.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a **containerized REST data-processing API** with automated CI/CD and validation checks, improving production reliability for operational workloads.

CPU Scheduling Simulator Code | *C++, React.js, JavaScript*
- Implemented **FCFS, SJF, SRT, RR, and MLQ algorithms** with interactive visualizations and metrics tooling, reducing debugging time by **20%**.